

PRODUCT REQUIREMENTS DOCUMENT

NetraJyoti MVP — Bengali voice-first eye-health guidance and referral support

Field	Value
Product	NetraJyoti AI
Document status	Capstone MVP requirements baseline + controlled GenAI pilot
Primary market	Rural Bengali-speaking communities in West Bengal
Product boundary	Guidance and referral support; not a diagnostic or emergency service
Version	5.1 1 August 2026

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Decision summary: Core MVP routing uses fixed, clinician-reviewable Bengali guidance and rules-based red-flag escalation. A controlled pilot may use GenAI only after the care outcome is fixed: to simplify a server-approved Bengali message or create a short caregiver summary from approved result content. It must not receive user inputs, change urgency, diagnose disease, prescribe treatment or invent service details.

1. Overview

NetraJyoti is a Bengali voice-first support system for rural West Bengal. It helps people describe an eye-health concern, understand a safe next step, and identify an appropriate type of eye-care service. It is designed for people who are more comfortable speaking Bengali than using text-heavy English applications.

The product exists to reduce delay, fear and confusion before treatment—not to replace clinicians. It provides a trusted first step when users may otherwise rely on family advice, informal care or forwarded messages.

2. Background and Current State

A rural Bengali-speaking person who notices an eye-health concern may first ask relatives, pharmacists, informal providers or rely on forwarded messages. They may not know whether the concern is urgent, where to travel, what services are available or which advice can be trusted. Text-heavy digital systems can add friction for people with low literacy, limited English confidence, shared devices, weak connectivity or low digital familiarity.

The current journey is therefore fragmented: concern → informal advice → delayed or uncertain decision → travel and service uncertainty. NetraJyoti is intended to create a safer first step before this uncertainty becomes avoidable delay.

3. Problem Statement

Many Bengali-speaking rural users do not know whether an eye problem is serious, where to go for help, or which advice to trust. Cost, travel, waiting time, poor communication and low digital confidence can make formal treatment harder. Delays can worsen avoidable eye-health problems and reduce timely treatment.

4. Evidence and Product Rationale

Evidence signal	Implication for the MVP
Sightsavers Sundarbans research (E1–E2) identifies treatment-seeking, cost, location, informal-first-contact and low-perceived-severity barriers.	Use a short Bengali-first flow, clear red-flag escalation and a referral direction instead of complex medical explanations.
Community Eye Health reporting (E3) describes unequal rural access and geographic difficulty in the Sundarbans.	Show the type of care to seek and support local service discovery; do not assume a hospital is nearby.
Misinformation and usability evidence (E4–E5, E10–E11) highlights trusted forwarded messages and friction in text-heavy systems.	Use concise voice/text prompts, explicit source and safety cues, caregiver sharing and low-reading-load interaction.
Comparator observations (E6–E9) identify waiting, coordination and service-experience concerns.	Set expectations conservatively and ensure service data is verified before public use.

Evidence Reference Key

The labels below identify the specific sources used in this PRD. E1–E5 provide contextual research evidence; E6–E11 are public comparator observations used to frame experience and accessibility risks; E12 identifies an existing West Bengal eye-health support reference. None of these references replaces local user research or clinical review.

Ref.	What the reference is	What it contributes to NetraJyoti
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Ref.	What the reference is	What it contributes to NetraJyoti
E1	Sightsavers Sunderbans Endline Survey (2018).	Context on treatment-seeking, financial and service-location barriers; supports referral guidance and the need to reduce uncertainty before travel.
E2	Sightsavers Sunderbans RMP Study (2018).	Context on informal first-contact care and low perceived severity; supports early red-flag prompts and non-judgemental Bengali guidance.
E3	Community Eye Health article: rural eye care in the Sundarbans.	Context on uneven access and geographic difficulty; supports local service-direction and does not assume a nearby hospital.
E4	Harvard Kennedy School Misinformation Review: WhatsApp misinformation in India.	General evidence that trusted messaging channels can spread uncertain information; supports clear sourcing and myth-busting as a future feature.
E5	World Health Organization note on the 'infodemic'.	Public-health context for harmful information overload; supports concise, careful and clearly bounded health communication.
E6	Public Disha Eye Hospitals complaint example.	Comparator signal for patient-experience risk such as coordination or behaviour concerns; supports honest expectation-setting before travel.
E7	Public Disha Eye Hospitals Justdial review page.	Comparator observations about waiting or service dissatisfaction; supports testing of care-journey expectations with local users.
E8	Public Susrut Eye Foundation Justdial review page.	Comparator observations about waiting and support experience; reinforces the need for clear and respectful referral information.
E9	Public Susrut Eye Foundation complaint-style review page.	Comparator observation of service-journey dissatisfaction; supports the decision not to overpromise service quality or availability.
E10	Digital India page describing DigiDrishti.	Comparator context on a multi-layer digital eye-care system; supports keeping the MVP flow short and guided for low-confidence users.
E11	Citizen Eyecare/DigiDrishti public app metadata.	Comparator signal on symptom-checking and language/usability demands; supports Bengali-first voice and text fallback testing.
E12	West Bengal Budget Speech reference to Chokher Alo eye-health support.	Shows relevant state-level eye-health support exists; future service direction should connect users to verified real programmes where appropriate.

Evidence limitation: the desk research supports the hypothesis but does not replace local primary research. Five consented JTBD interviews and partner validation are required before a real pilot.

5. Terminology

Term	Meaning in this PRD
Red flag	A predefined response requiring urgent eye-care guidance, including sudden blurred vision, severe pain, injury or chemical exposure.
Routine guidance	Fixed, non-diagnostic next-step guidance for concerns without configured urgent signals.
Human escalation	Direction to consult an ASHA worker, qualified health worker, clinic or eye-care provider when the app cannot safely classify a concern.
Service directory	A list of eye-care services. MVP entries are demo-only until partner verified.
ASHA worker	Accredited Social Health Activist/community-health worker who may assist a user but does not diagnose through NetraJyoti.
Demo-only data	Prototype information that must not be relied on for travel or treatment decisions.

6. Goals, Product Hills and Non-Goals

Goals

- Enable a rural Bengali-speaking user to describe a concern through voice or text without needing English typing.
- Route red-flag responses to immediate urgent-care guidance using fixed safety rules.
- Give non-diagnostic, simple Bengali next-step guidance and an appropriate service type.
- Help caregivers and ASHA/community-health workers support a user without acting as a clinician.
- Create a measurable, clinically reviewable capstone MVP for usability and referral testing.

Product Hills

- A rural Bengali-speaking user can understand the next safe eye-care action without English typing or medical knowledge.
- A caregiver can confidently share the next step without interpreting medical information or making a diagnosis.
- An ASHA/community-health worker can assist a villager through a repeatable flow without being expected to prescribe or diagnose.
- No urgent configured red-flag case is delayed by generated wording, unclear navigation or an unverified service listing.

Non-Goals

- Diagnosing eye disease, interpreting images, prescribing medicines or replacing professional eye care.
- Autonomous appointment booking, medical-record integration or payment processing.
- Publishing unverified provider details, camp dates or emergency availability.
- Using GenAI for triage, diagnosis, prescriptions, free-form clinical chat or altering approved health guidance.
- Expanding beyond Bengali or rural West Bengal during the MVP stage.

7. Target Users and Jobs-to-be-Done

Persona	Primary job	Key barrier	MVP value
Maya — rural user, 58	Decide what to do when vision changes or the eye is uncomfortable.	Low literacy, fear of cost, English typing and reliance on relatives.	Can speak Bengali and receive short, safe next-step guidance.
Rahim — caregiver, 32	Help a family member decide and prepare for travel.	Uncertain advice and difficulty coordinating care from another location.	Gets a shareable plain-language summary.
Soma — ASHA/community-health worker	Support a villager without giving clinical advice.	Limited time and need for trusted referral guidance.	Uses a guided, non-diagnostic assisted flow with escalation.

Persona	Primary job	Key barrier	MVP value
Outreach coordinator	Keep users aware of appropriate care options.	Outdated service and camp information causes failed referrals.	Maintains partner-verified listings in a later pilot phase.

Persona Profiles and Design Implications

The following provisional personas are grounded in the problem hypothesis and desk research. They must be tested and refined through the required real JTBD interviews; they are not substitutes for primary evidence.

Maya — rural Bengali-speaking user

Context: Maya is an older adult in a rural area who notices a change in vision or eye discomfort. She may use a shared phone, have low confidence reading long text, and depend on relatives for travel or decisions.

- Need: a calm, short answer in Bengali that explains the next safe action without medical jargon.
- Behaviour: may seek family, pharmacy or informal advice before formal care because travel, cost and uncertainty feel high.
- Design requirements: large Bengali labels, one decision at a time, voice input with text fallback, visible urgent warning and no English-only gate.

Rahim — caregiver and decision supporter

Context: Rahim may help a parent or relative from another household or village. He needs enough information to coordinate travel, explain the next step and reduce anxiety without becoming responsible for medical decisions.

- Need: a concise shareable summary that states the urgency and recommended care type.
- Behaviour: compares advice from relatives, local providers and online messages; needs a credible, simple source before asking family to act.
- Design requirements: copy/share action, plain-language outcome, clear disclaimer and service data that is visibly demo-only until verified.

Soma — ASHA or community-health worker

Context: Soma supports several community members with limited time. She needs a repeatable way to help a villager navigate a concern while staying within her non-clinical role.

- Need: an assisted, explainable flow with immediate escalation when uncertainty or red flags occur.
- Behaviour: may complete the flow together with a user and explain the result in familiar language.
- Design requirements: short prompts, fixed safety rules, human-escalation route, no diagnostic label and no expectation that she updates clinical content.

Outreach coordinator — service-information steward

Context: The coordinator works with eye camps, vision centres or hospitals and understands that listing errors can cause avoidable travel and missed care.

- Need: a controlled process to add, update and expire location, contact and service information.
- Behaviour: confirms data with partner organisations rather than relying on user-submitted information.
- Design requirements: future authenticated workflow with source, reviewer, last-reviewed date and expiry. This is explicitly post-MVP; the public MVP uses demo-only entries.

8. Comparative Analysis, Competition and Differentiation

Alternative / competitor	Observed limitation or role	NetraJyoti differentiation
Informal advice: relatives, pharmacies and forwarded messages	Often trusted but may be incomplete, inconsistent or not linked to eye-care services.	Bengali-first safety boundary, fixed next-step guidance and referral direction.
Eye hospitals, camps and vision centres	Provide care but can be difficult to discover or navigate before travel.	Helps the user identify the appropriate type of care before travel; does not claim to replace providers.
DigiDrishti and other text-heavy digital systems	Can require reading confidence, navigation across multiple steps and familiarity with digital health flows.	Short, voice-first Bengali interaction with visible text fallback and caregiver support.
General AI/chat tools	May produce unverified or diagnostic-looking health answers.	Rules-based urgency and fixed clinician-reviewable wording. A controlled pilot may simplify only server-approved copy; it never generates clinical advice.

Competitive position: NetraJyoti is not a hospital or diagnostic platform. Its differentiated role is a trustworthy, Bengali-first first step between informal advice and verified formal eye care.

9. MVP Scope and Feature Catalogue

Feature catalogue

Priority	Features
Must have	Bengali voice input; text fallback; consent; rules-based red-flag routing; fixed Bengali guidance; non-diagnosis disclaimer; demo-only service direction; caregiver summary.
Controlled pilot only	AI-assisted plain-Bengali explanation and short caregiver summary from a server-approved result message, behind a feature flag, with clinical, privacy and output-evaluation gates.
Should have	Feedback capture; ASHA-assisted workflow; verified-directory management; eye-camp reminders after partner validation.
Could have	Offline content cache; WhatsApp sharing; partner-approved appointment-request support; clinician-approved FAQ knowledge base.
Won't have in MVP	Disease or image diagnosis; prescriptions; autonomous booking; medical-record integration; publicly relied-upon unverified provider data.

In scope	Deferred / out of scope
Bengali voice input with text fallback; Bengali text and spoken output.	Image-based or disease diagnosis.

In scope	Deferred / out of scope
Rules-based urgency checks for sudden blurred vision, severe pain, injury and chemical exposure.	Prescriptions, treatment plans or dosage advice.
Fixed Bengali guidance for urgent, routine and human-escalation paths; controlled-pilot simplification of an approved result message only.	Generative AI health answers, free-form clinical chat, diagnosis, prescriptions or AI-driven triage.
Demo-only service directory; fixed caregiver summary; controlled-pilot AI-assisted caregiver summary from approved result content; feedback capture; safety and privacy consent.	Verified public directory, automated booking, WhatsApp integration, payment or EHR connectivity.

10. Required User Stories and Acceptance Criteria

ID	User story	Acceptance criteria
US-01	As a rural Bengali-speaking user, I want to describe my eye problem by voice so I do not need to type or read English.	The user can start Bengali speech input; recognised text appears for review; text entry remains available if voice input fails; the app states that voice capability depends on the device/browser.
US-02	As a user with sudden vision loss, severe pain, eye injury or chemical exposure, I want immediate advice to seek emergency care.	Any configured red flag returns an URGENT result; the screen and audio say not to wait; no diagnosis or medication recommendation is shown; relevant care types are suggested.
US-03	As a user with a non-urgent concern, I want simple Bengali information on what to do next without being told I have a disease.	Routine cases receive fixed, simple Bengali guidance; the disclaimer is visible; response language contains no diagnosis, prescription or unsupported disease claim.
US-04	As a caregiver, I want a short referral summary so I can help my family member travel and prepare.	A Bengali summary can be copied or shared using supported device sharing; it includes the next step and disclaimer but no diagnosis.
US-05	As an ASHA worker, I want an assisted flow so I can guide a villager safely without acting as a clinician.	The same guided flow can be completed together; uncertainty routes to human escalation; the interface consistently states that it is not diagnosis.
US-06	As an outreach coordinator, I want to update eye-camp dates and locations so users receive current service information.	Not in the public MVP UI. A future authenticated partner workflow must record source, reviewer, last-reviewed date and expiry; unverified entries must never be marked verified.
US-07	As a Bengali-speaking user, I want an approved next-step message explained in simpler Bengali so I can understand it easily.	Controlled pilot only. Rules select the outcome and approved message before any model call. The model receives only approved copy, cannot change urgency or provide clinical advice, and its short output is labelled AI-assisted. On timeout, invalid output or unavailable configuration, show the original approved copy.
US-08	As a caregiver, I want a short plain-Bengali summary of an approved result so I can help my family member act on the next step.	Controlled pilot only. The backend sends only the already-approved result content to create a brief caregiver summary. It cannot change urgency, care type or clinical facts; it is labelled AI-assisted. Do not send voice transcripts, symptom details, optional questions, district/location or identifiers. On timeout, unsafe output or unavailable configuration, show the fixed approved caregiver summary.
US-09	As a NetraJyoti user, I want clear consent and privacy controls so I understand how my information is	Jira NET-8. Before continuing, acknowledge the versioned safety/privacy notice. Optional collection for research recording, referral sharing or analytics

ID	User story	Acceptance criteria
	handled.	requires separate consent; the no-storage core guidance path remains available.
US-10	As the product team, we want safety, accessibility and fallback checks so the MVP remains dependable for Bengali-speaking users.	Jira NET-9. Test all URGENT, ROUTINE and HUMAN ESCALATION rules; no-diagnosis wording; Bengali readability; mobile touch targets; voice, audio, network and directory fallbacks; and the agreed device/browser matrix.
US-11	As the product team, we want privacy-safe pilot feedback and analytics so we can measure usability without unnecessary data collection.	Jira NET-10. After analytics consent, record only the approved aggregate event schema. Exclude voice, symptoms, free text, identifiers, precise location and raw GenAI output; support opt-out and documented retention/access controls.

11. Functional Requirements

ID	Area	Requirement
FR-01	Start and consent	Show the non-diagnosis and emergency disclaimer before the assessment; require acknowledgement of the privacy statement before continuing.
FR-02	Concern capture	Accept only approved concern categories: blurred vision, pain, redness/watering, injury, chemical exposure or other. Reject invalid API values.
FR-03	Safety rules	Return URGENT when injury/chemical exposure, severe pain, or sudden onset of blurred vision is recorded. Return HUMAN ESCALATION for 'Other'. Otherwise return ROUTINE.
FR-04	Guidance output	Use fixed Bengali copy maintained in the rules service. Read it aloud where browser speech synthesis is available; always keep text visible.
FR-05	Service direction	Recommend a type of care based on urgency. Show the MVP directory only as DEMO ONLY and tell the user to verify locally before travel.
FR-06	Caregiver summary	Allow a concise, shareable summary containing result title and guidance; use browser-native sharing or copy as fallback.
FR-07	Feedback	Capture a clarity/usefulness rating and optional comment in a future persistent store. The prototype may acknowledge receipt without retaining data.
FR-08	Controlled GenAI simplification	After the rules engine has selected an outcome, a server-side feature flag may send only the approved Bengali message to an approved GenAI provider for plain-language simplification. Do not send voice transcripts, symptom details, optional questions, district/location or identifiers. Validate output; label it AI-assisted; fall back to the original approved copy on error, timeout, unsafe output or missing configuration.
FR-09	Controlled AI-assisted caregiver summary	After the rules engine has selected the outcome, a server-side feature flag may create a concise caregiver summary only from the already-approved result content. Preserve the outcome, urgency, care type and original guidance; do not add facts, diagnoses, medicines or provider details. Label the accepted summary AI-assisted and fall back to the fixed approved caregiver summary on error, timeout, unsafe output or missing configuration. Do not send user-entered data to the provider.

Requirements Traceability

PRD requirement	Jira delivery item	Status
FR-08 — Controlled GenAI simplification	NET-11 — AI-assisted plain-Bengali explanation of approved guidance	Controlled-pilot story
FR-09 — AI-assisted caregiver summary	NET-12 — AI-assisted caregiver summary from approved result content	Controlled-pilot story (3 points)

12. Safety, Privacy and Ethical Requirements

Safety requirement: Urgency is determined only by the reviewed rules engine. The public MVP has no GenAI clinical guidance. A controlled GenAI pilot may simplify a server-approved Bengali message only after the outcome is fixed; it must be independently evaluated and prevented from changing urgency, diagnosing, prescribing or adding facts.

- Display 'Guidance, not diagnosis' and urgent-care instructions throughout the flow.
- Collect the minimum data needed: concern, urgent flags, district and optional question. Do not request medical records, photographs, identity numbers or payment information.
- Do not store the optional question in the MVP; do not send it to an external AI service.
- For the controlled pilot, send only approved guidance copy from the server; never send voice transcripts, symptom details, optional questions, district/location or identifiers to the GenAI provider.
- Keep the GenAI provider key in server-side configuration only. Label any accepted simplified copy as AI-assisted and display the original approved message when the service fails or output does not pass validation.
- The controlled-pilot caregiver summary may use only already-approved result content. It must retain the fixed summary as fallback and must not receive user-entered data or create clinical advice.
- Reject GenAI output if it contains a diagnosis or disease name, medicine/dosage, changed urgency or care type, unsupported clinical facts, or new service/provider detail. Reject output that is not concise Bengali plain language; show the approved fixed copy instead.
- Use only partner-verified service data for any pilot or public launch; display source, reviewer and last-reviewed date in the partner system.
- Provide a human escalation option for uncertainty, repeated failure, urgent conditions and user request.
- Obtain informed consent before research recordings; anonymise interview records and secure recordings separately from study notes.

Controlled-GenAI Operational Ownership

Responsibility	Named role / requirement
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Responsibility	Named role / requirement
Feature-flag and provider-key owner	Backend owner; enables the pilot only after release gates and keeps the key in server-side configuration.
Approved Bengali source-content owner	Qualified clinical/content reviewer; owns approved messages, caregiver templates, review date and expiry cadence.
Output-evaluation owner	Product and clinical reviewers; retain non-content evaluation evidence for faithfulness, prohibited-content rejection and fallback behaviour.
Incident owner	Pilot operations owner; disables the feature flag, preserves the fixed-copy path and routes safety concerns to the defined escalation process.
Technical logging policy	Record only non-content audit metadata: selected outcome, feature-flag state, validation result and fallback reason. Do not log user health content, location or identity.

13. Non-Functional Requirements

Area	MVP requirement
Accessibility	Mobile-first layout; Bengali-first labels; touch targets; voice and text alternatives; readable contrast; no mandatory English input.
Performance	Show a response or safe connection-error message within 10 seconds on ordinary mobile connectivity; never leave the user without an urgent-care instruction.
Reliability	If voice input, sharing or service lookup fails, keep text guidance visible and present a practical fallback.
Security	HTTPS in deployment; API input validation; server-side configuration only; no secret keys in the browser; rate limiting before public pilot.
Controlled GenAI	Feature flag defaults off; model receives approved copy only; bounded Bengali output is evaluated and validated; original approved copy is the safe fallback for all failures.
Maintainability	Rules, Bengali wording and service listings must be versioned and reviewable; safety changes require clinician approval.

14. User Flow

1. User opens NetraJyoti and sees the Bengali-first purpose and non-diagnosis boundary.
2. User acknowledges safety and privacy information.
3. User chooses a concern, records urgent flags, enters a district and optionally speaks or types a question.
4. Rules engine classifies the session as URGENT, ROUTINE or HUMAN ESCALATION.
5. User receives fixed Bengali text and optional audio. Urgent cases are told not to wait.
6. In the controlled pilot only, the user may request an AI-assisted plain-Bengali explanation of that already-approved message; the fixed message remains visible and is the fallback.
7. User sees a demo-only service direction, may share a caregiver summary, and can provide feedback or restart.

15. Success Metrics and Pilot Validation

Metric	Pilot measure	Initial success threshold
Flow completion	Percentage of consented users completing concern capture and receiving a result.	≥70% with assisted onboarding documented separately.
Red-flag routing	Percentage of scripted red-flag test cases routed to URGENT.	100% in test suite and clinical review.
Comprehension	User can repeat their next step in their own words after task completion.	≥80% of usability-test participants.
Voice usability	Users who can successfully use voice or switch to text without abandoning the flow.	≥75%; segment by device/noise/assistance.
Directory accuracy	Partner-confirmed availability and contact details.	100% before any non-demo listing is shown.
Trust and safety	Users reporting that guidance was clear; unresolved/unsafe sessions flagged for review.	No unresolved urgent test case; qualitative issues logged.
Caregiver-summary comprehension	Caregiver can repeat the approved next step after reading or hearing the AI-assisted summary.	≥80% in controlled-pilot usability sessions.
Outcome fidelity	Sampled GenAI summaries preserve the selected urgency, care type and next-step instruction.	100% in evaluated test corpus.
Safety fallback	Rate of rejected/failed GenAI requests that display the approved fixed copy; verify no user-entered data reaches the provider.	100% safe fallback; 0 user-data transfers in test evidence.

16. Testing and Release Gates

8. Run unit tests for all red-flag, routine and human-escalation paths; reject invalid concern values.
9. Conduct clinical review of safety rules, Bengali scripts and disclaimers with an ophthalmologist or authorised eye-care partner.
10. Conduct task-based usability sessions with rural users, caregivers and ASHA/community-health workers; test comprehension, voice recognition and fallback text entry.
11. Validate every service entry, contact route, hours and camp date with a partner before moving from DEMO ONLY to pilot data.
12. Test slow networks, older devices, microphone denial, browser voice unavailability, empty location and service lookup failure.
13. Complete privacy review, deploy HTTPS, protect configuration and add operational monitoring/rate limiting before any public pilot.
14. Before enabling the controlled GenAI pilot, obtain clinician sign-off on source messages, privacy approval, prompt/output evaluation evidence, explicit prohibited-content validation tests, non-content audit logging checks and a tested fallback to the fixed Bengali message and caregiver summary.

17. Dependencies, Risks and Decisions

Dependency / risk	Mitigation or decision
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Dependency / risk	Mitigation or decision
Clinical safety review is not yet evidenced as complete.	Do not launch publicly; obtain documented review and periodic ownership before pilot.
Service data may be stale or unavailable.	Keep demo-only warning now; add authenticated, partner-owned update and expiry workflow later.
Speech quality varies by device, accent and noise.	Use browser speech only as an assistive input; provide text fallback and test with target users.
Local research is incomplete.	Conduct five real, consented JTBD interviews; retain raw notes, exact quotes and recording references; do not fabricate findings.
Users may interpret guidance as diagnosis.	Repeat boundary language at consent, result and share points; avoid disease names and treatment recommendations.
GenAI simplification could introduce an unsafe or misleading statement.	Keep it feature-flagged and pilot-only; provide only approved source copy; validate bounded output; label it AI-assisted; preserve the original approved copy and fall back to it on any failure.
A GenAI provider credential could be exposed or unavailable.	Configure the provider key only in the backend environment; never in React/browser code; disable the feature when configuration is absent and retain fixed-copy guidance.

18. Future Roadmap (Post-MVP)

- Authenticated coordinator tools for verified service, camp and referral updates.
- Clinician-approved Bengali knowledge base, retrieval controls and safety evaluation for limited non-diagnostic FAQs.
- Expand the controlled AI-assisted plain-language pilot only after documented comprehension, privacy, safety and fallback results; do not broaden it into clinical chat without a separate PRD and governance review.
- ASHA-assisted mode, reminder workflow and an offline content cache following field validation.
- Partner-approved referral follow-up and aggregate analytics with consent and minimum-data safeguards.
- Broader district coverage only after local evidence, service partnerships and safety governance are established.

Next-Phase GenAI Capability Candidates

The following are product opportunities, not committed MVP functionality. Each requires a separate backlog story, clinician-approved source content, privacy review, output evaluation, feature flag, human-escalation route and safe fixed-copy fallback before pilot use.

- Explain clinician-approved eye-health FAQs in simpler Bengali without adding clinical facts.
- Provide voice-led clarification of approved referral and visit-preparation instructions.
- Create a concise Bengali caregiver summary only from the already-approved result content.
- Help an ASHA/community-health worker navigate approved, non-diagnostic guidance during an assisted session.
- Recognise that a request is unclear or outside the approved content and direct the user to a qualified health worker instead of attempting an answer.

Explicit boundary: These next-phase candidates must never diagnose, interpret images, prescribe, classify urgency, replace the reviewed rules engine, or generate ungrounded health advice.

18A. Future AI and Agentic Capabilities (Post-MVP)

The following capabilities are a future roadmap, not MVP commitments. They may be introduced only after clinical/content review, privacy impact assessment, Bengali usability validation, partner data governance and measured pilot performance.

Core safety principle: clinician-reviewable deterministic rules remain the sole mechanism that selects URGENT, ROUTINE or HUMAN SUPPORT. AI, voice input, free text, location and patient details cannot alter the route.

Capability	Future purpose	Required controls	Non-negotiable boundary
Voice and language intelligence	Improve Bengali speech recognition, text-to-speech and plain-language simplification.	Use reviewed Bengali content, usability testing and a clear text/manual fallback.	Cannot determine urgency or diagnose.
Verified service-directory retrieval assistant	Find nearby vision centres, clinics, PHCs, eye camps or ASHA support from governed directory data.	Use partner-approved records, verification date and source metadata.	Cannot invent services or make bookings autonomously.
Coordinator content assistant	Draft Bengali eye-camp updates, reminders and outreach messages.	Human review and approval are required before publication.	Cannot publish, alter dates or contact users on its own.
Service-data quality assistant	Flag missing, expired, duplicate or inconsistent directory records.	Create a review queue with an audit trail.	Cannot auto-correct or publish data.
Privacy-safe analytics assistant	Summarise anonymous journey friction, drop-offs and support needs.	Use consented, aggregated and de-identified events only.	Cannot access raw voice, free text, identifiers or precise location.
Governed conversational support	Clarify app navigation and explain reviewed guidance after the route is fixed.	Use bounded prompts, approved knowledge and human hand-off for uncertainty.	Cannot change the route, prescribe, diagnose or give emergency reassurance.

Sequencing: first expand Bengali voice and controlled caregiver summaries; next establish verified service retrieval and coordinator workflows; then introduce data-quality and aggregate analytics assistance. Any agentic workflow must be assistive, bounded, reviewable and reversible—not autonomous.

19. Appendix

Appendix	Contents
A	Evidence register E1–E12 and source links in NETRAJYOTI.docx.
B	JTBD interview script, consent wording, raw-notes template and recording inventory in the Phase 2 submission pack.
C	Safety escalation matrix, clinical-review gate and service-data verification requirements.
D	NetraJyoti MVP wireframe, user flow and Java/React technical implementation.
E	Test cases, pilot metrics and release-gate checklist.

20. Source Basis

This PRD synthesises the project summary and evidence appendix in NETRAJYOTI.docx and the Phase 2 strategy, interview plan, safety matrix, MVP plan and user stories in NetraJyoti_Capstone_Project_Submission_Pack_v2.docx. Evidence references E1–E12 and comparator materials remain in the source documents. This PRD does not claim that the required five real interviews or clinical validation have already been completed.